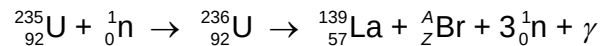


- 8 (a)** In a nuclear reaction, a uranium-235 ($^{235}_{92}\text{U}$) nuclide is transformed into an unstable uranium-236 nuclide ($^{236}_{92}\text{U}$) through bombardment by a slow-moving neutron. The unstable uranium-236 nuclide undergoes nuclear fission to form stable products of a lanthanum-139 nuclide ($^{139}_{57}\text{La}$) and a nuclide of bromine (^A_ZBr).



Data for the nuclei in the reaction are given in Fig. 8.1.

nucleus		mass / <i>u</i>
proton	${}_1^1\text{p}$	1.00728
neutron	${}_0^1\text{n}$	1.00866
lanthanum	${}_{57}^{139}\text{La}$	138.906
Uranium	${}_{92}^{235}\text{U}$	235.044

Fig. 8.1

- (i)** Determine the nucleon number *A* and proton number *Z* of the bromine nuclide.

A =

Z = [1]

- (ii)** Calculate the binding energy of the uranium-235 nuclide to 4 significant figures.

binding energy = MeV [3]

(iii) State the feature of this equation that indicates a chain reaction may be possible.

.....
 [1]

(b) In a laboratory source of strontium-90, the number of atoms present in the year 2013 was 2.36×10^{13} . Strontium-90 decays by emission of a β -particle and this nuclide has a half-life of 28 years.

(i) State what is a β -particle.

..... [1]

(ii) Calculate the activity of the source in the year 2113.

activity = Bq [2]

Section B

Answer **one** question from this Section in the spaces provided.

9. (a) Define simple harmonic motion.